NOVEL ELECTROSPINNING PVDF/BI_2O_3/CEO_2 COMPOSITE YARN: A VERSATILE APPROACH TOWARDS FUNCTIONAL X-RAY SHIELDING TEXTILE

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ABSTRACT

Recent environmental challenges, particularly concerns regarding nuclear waste management, have heightened the urgency for developing practical radiation protection solutions suitable for daily use [1]. While existing nanofiber membranes exhibit promising radiation shielding capabilities, their inherent limitations in flexibility and processability significantly restrict their integration into wearable applications [2]. Traditional lead-based protective materials, despite their effectiveness, present substantial weak points in terms of weight and mobility constraints [3].

This study presents a novel approach to developing functional X-ray shielding textiles through Electrospinning PVDF/Bi_2O_3/CeO_2 composite yarns [4]. With increasing concerns about nuclear waste management and radiation protection in daily applications, there is an urgent need for lightweight, flexible protective materials that overcome the limitations of traditional lead-based shields. Considering the heavyweight of conventional lead garments, PVDF was selected as the polymer matrix due to its lightweight nature and versatile wearable applications. This research introduces an innovative composite system incorporating Poly(vinylidene fluoride) (PVDF) with high-Z elements, specifically Bismuth Oxide (Bi_2O_3) and Cerium Oxide (CeO_2) nanoparticles, to achieve an optimal balance between protection efficacy and textile functionality [5].

Characterization using X-ray computed tomography (XCT), scanning electron microscopy (SEM/EDX), and Fourier transform infrared spectroscopy (FTIR) demonstrates that the thick PVDF-Bi_2O_3/CeO_2 composite achieves comparable X-ray shielding performance to commercial lead films. Tensile fibre test evaluates the yarn's strength. This analysis confirms an enhanced β phase content, which contributes to improved piezoelectric properties. This versatile approach demonstrates promising potential for developing next-generation, multifunctional protective textiles that combine both radiation shielding and piezoelectricity. Furthermore, the electrospun yarn formation of PVDF-Bi_2O_3/CeO_2 requires better optimization of material ratios and processing parameters to achieve sufficient X-ray shielding effectiveness for textile applications.

1 INTRODUCTION

The growing concerns over radiation exposure in medical, industrial, and nuclear environments have intensified the demand for comfortable personal protective equipment. Traditional lead-based shielding materials, while demonstrating excellent X-ray attenuation capabilities, present significant limitations, including excessive weight, reduced mobility, and potential health risks associated with lead exposure. These constraints have motivated researchers to explore alternative lightweight materials that can provide comparable protection without compromising user comfort and wearability [6].

Recent advances in polymer nanocomposites have opened new avenues for developing functional protective textiles. Among various polymer matrices, Poly(vinylidene fluoride) (PVDF) emerges as a promising candidate due to its exceptional mechanical flexibility, chemical stability, and inherent

piezoelectric properties. The incorporation of high atomic number (high-Z) elements such as bismuth oxide (Bi_2O_3) and cerium oxide (CeO_2) into PVDF matrices offers a strategic approach to enhance X-ray shielding performance while maintaining textile processability [4].

Remarkably, this material combination creates a dual-beneficial relationship where the added metal oxides not only provide radiation attenuation but also simultaneously enhance PVDF's piezoelectric characteristics through increased β-phase crystallinity. This synergistic effect enables the development of multifunctional smart textiles with both protective and energy harvesting capabilities. This study presents the fabrication and characterization of electrospun PVDF/Bi_2O_3/CeO_2 composite fibers, demonstrating their potential as next-generation lightweight radiation shielding textiles [5].

2 METHODOLOGY

2.1 Material Synthesis

The composite was developed using PVDF nano powder (Mw = 534,000, Sigma-Aldrich) as the polymer, integrated with high-Z elements Bi_2O_3 and CeO_2 (Sigma-Aldrich) as X-ray attenuating materials. This combination leverages PVDF's mechanical flexibility and chemical stability, while the Bi_2O_3 and CeO_2 provide effective X-ray attenuation. The solvent comprised a 3:1 ratio of dimethyl sulfoxide (DMSO) to acetone.

The polymer solution was prepared by mixing DMSO and acetone (3:1) with mechanical stirring. PVDF nano powder was gradually added to the solvent three times, with ultrasonication used to ensure it was well dispersed. The resulting solution achieves transparency and clarity. This prepared solution was divided into two equal portions. One portion was pure PVDF solution, while the other was incorporated with Bi_2O_3 and CeO_2. The composite mixture was also processed by ultrasonication until a uniform dispersion was achieved. The electrospinning process was conducted under a voltage of 15 kV, a tip-to-collector distance of 17 cm, and a solution feed rate of 0.6 mL/h. Fiber deposition was performed onto aluminum foil collectors, PVDF, and PVDF-Bi_2O_3-CeO_2 composite.

2.2 Yarn and fabrication

Both pure PVDF yarn and High Z elements yarn's fabrication process was conducted in collaboration with Nayi Instrument Technology Co., Ltd, China, utilizing their specialized electrospinning equipment for continuous yarn production. To achieve a better formation of electrospinning yarn, the solution was chosen to use DMF instead of using DMSO, and Bi_2O_3 and CeO_2 were selected with a smaller particle size of 20 nm. Based on the optimized formulation from preliminary experiments, the material composition was precisely controlled at a mass ratio of 1g PVDF to 0.35g total metal oxides, specifically comprising 0.25g Bi_2O_3 and 0.1g CeO_2. Both metal oxide nanoparticles were selected to have an average particle size of 20 nm, ensuring optimal dispersion and functional performance.

The electrospinning yarn production employed a positive high voltage of 8.6 kV and a negative voltage of 8.8 kV. The solution flow rate was adjusted to 1.5ml/h. With overnight observation of the yarn funnel collector in a steady state for producing electrospinning yarn, the temperature is controlled in 50℃ and the humidity is 24.5%.

2.3 Materials Characterization

The morphological, structural, and functional characteristics of the fabricated membranes were evaluated and conducted in multiple analytical techniques: X-ray computed tomography (XCT) for X-ray attenuation, scanning electron microscopy (SEM) for morphology analysis, and Fourier transform infrared spectroscopy (FTIR) for β phase identification.

Following the yarn fabrication, tensile fibre testing was conducted to assess the mechanical properties and structural integrity of both pure PVDF and composite PVDF-Bi_2O_3/CeO_2 yarns, which provides the impact of these metal oxides' incorporation on yarn mechanical performance.

3 RESULT AND DISCUSSION

3.1 Morphological Characterization

SEM Analysis of Materials Membrane

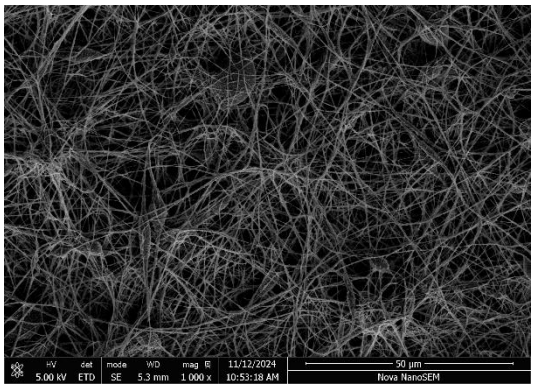


Figure 1 PVDF Membrane

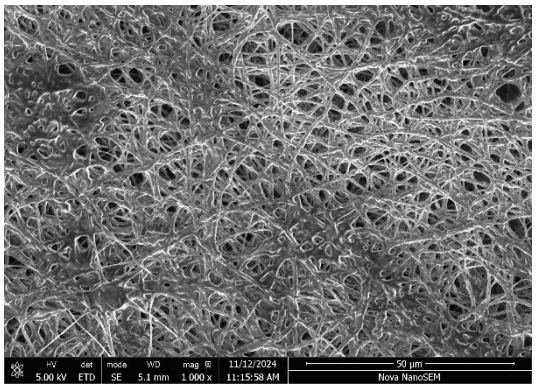


Figure 2 PVDF-Bi_2O_3/CeO_2 Membrane

Morphological analysis via scanning electron microscopy elucidates the structural characteristics of the fabricated fibrous networks at 1000 magnification. The pristine PVDF architecture exhibits well-defined nanofibrous formations with consistent dimensional uniformity, while the PVDF matrices containing Bi_2O_3 and CeO_2 display marginally enhanced fiber diameters attributed to the successful incorporation of high-Z elements.

SEM Analysis of Yarn Formation

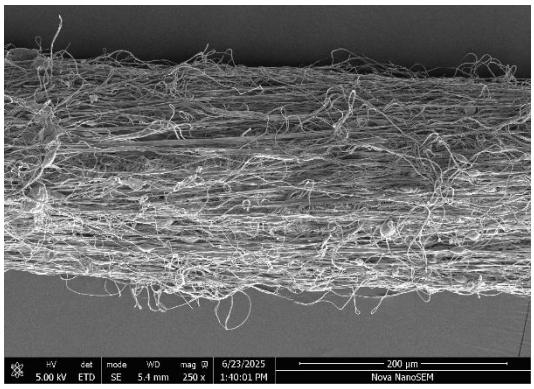


Figure 3 PVDF Yarn

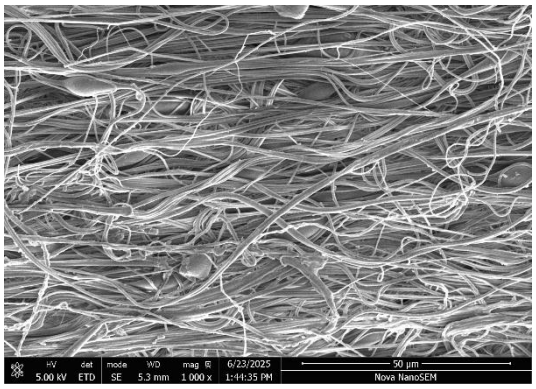


Figure 4 PVDF Yarn Twisting Structure

SEM of pure PVDF yarn (also processed by Nayi) analysis reveals the yarn's twisted configuration at 250 magnifications, with occasional bead formations present for structural balance. At 1000 magnification, individual fibres demonstrate consistent directional alignment attributed to the electrospinning process.

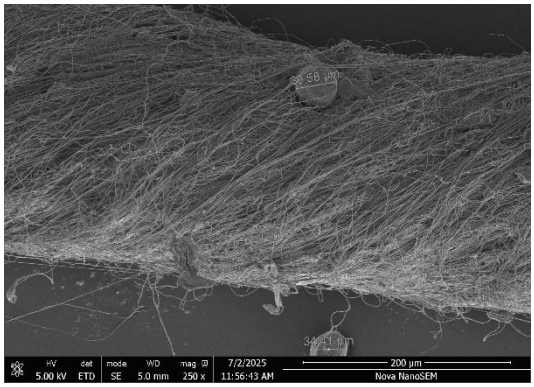


Figure 5 PVDF-Bi_2O_3/CeO_2 Yarn

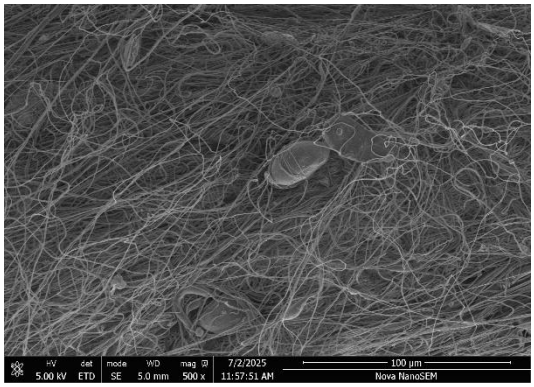


Figure 6 PVDF-Bi_2O_3/CeO_2 Yarn Twisting Structure

These SEM images show larger bead formation in the high-Z composite yarn, although these exhibit a clustered, agglomerated morphology rather than the typical spherical shapes of PVDF. It indicates metal oxide accumulation. However, the yarn demonstrates excellent structural integrity with clearly defined twisting pathways remaining well-preserved throughout.

Energy-dispersive X-ray spectroscopy (EDX)

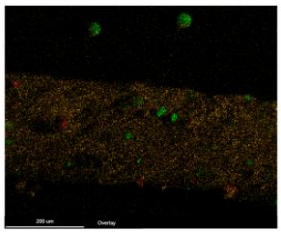


Figure 7 PVDF-Bi2O3/CeO2

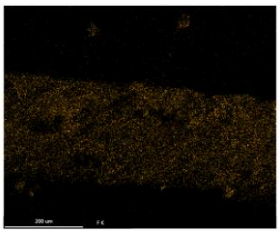


Figure 8 PVDF

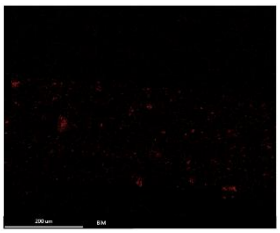


Figure 9 Bi2O3

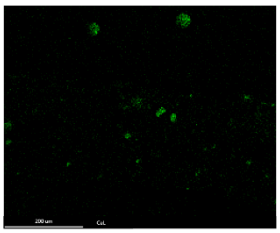


Figure 10 CeO2

The EDX mapping reveals the dispersion of high-Z elements dispersion within the PVDF composite. The three different colors in mapping demonstrate the PVDF Matrix with Bi_2O_3 and CeO_2. However, the dispersion during the solution preparation stage is not perfect, and the metal oxide material nanoparticles are still easy to secondary agglomeration.

3.2 Structural Properties

Membrane & Yarn in FTIR

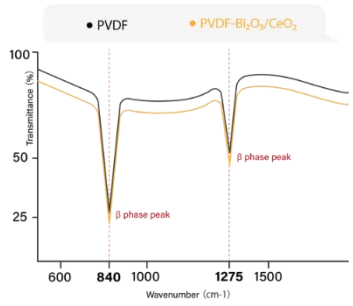


Figure 11 FTIR Spectra Analysis: β Phase Identification in PVDF Materials

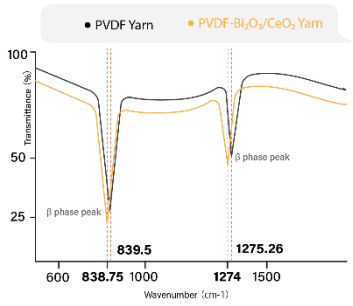


Figure 12 FTIR Spectra Analysis: β Phase Identification in PVDF Electrospun Yarn

Both sample types (PVDF and PVDF-Bi_2O_3/CeO_2) exhibit characteristic β-phase formation. The FTIR spectra clearly show characteristic β-phase peaks at 840 cm^{-1} and 1275 cm^{-1}, with enhanced intensity in the PVDF-Bi_2O_3/CeO_2 composite indicating increased β-phase content.

3.3 X-ray Shielding Performance

X-ray Shielding Effective in Membrane

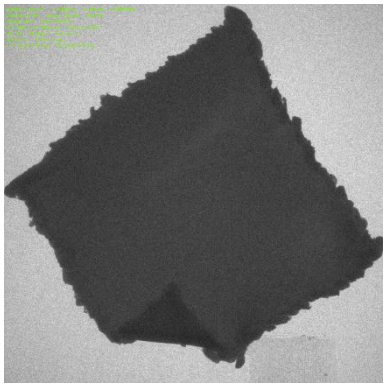


Figure 13 Comparison-Lead-commercial

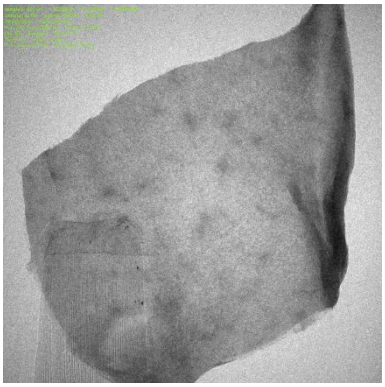


Figure 14 Composite-PVDF-Bi_2O_3/CeO_2 thin

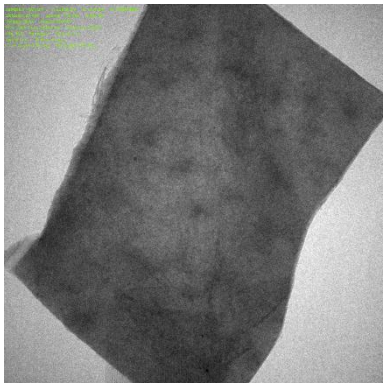


Figure 15 Composite-PVDF-Bi_2O_3/CeO_2 thick

These images are captured by ZEISS Xradia 410 Versa. By comparing it to the lead film, this is conducted on three samples with the same X-ray energy parameters, 80 kV-10.00 W, revealing the relative attenuation capabilities through grayscale contrast. The commercial lead film exhibits maximal attenuation (darkest contrast), and the thick PVDF-based composite achieves comparable X-ray shielding performance.

X-ray Shielding Effective in Yarn

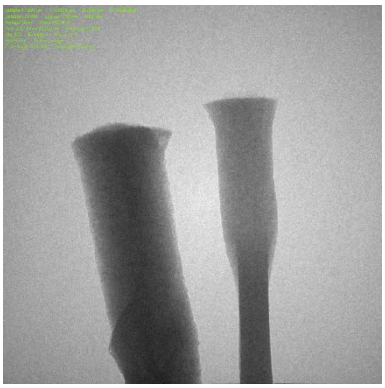


Figure 16 Composite-PVDF-Bi_2O_3/CeO_2 thick membrane in hand twist

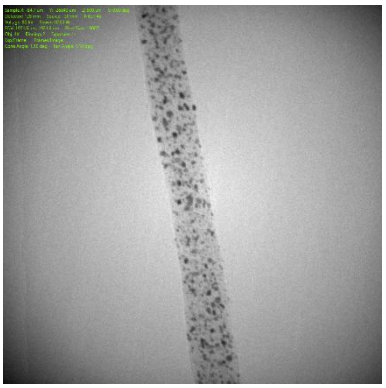


Figure 17 Electrospun Composite-PVDF-Bi_2O_3/CeO_2

Under the same X-ray condition, comparative analysis between hand-twisted thick membranes and electrospun yarns reveals notable differences in shielding performance. The electrospun yarns demonstrated stable and consistent characteristics, but a lower density of metal oxide nanoparticles.

3.4 Mechanical Properties of Yarns

Tensile for PVDF Yarn and PVDF-Bi_2O_3/CeO_2 Yarn

Tensile testing of twenty samples each (pure PVDF and PVDF-Bi_2O_3/CeO_2 yarns) using the BS EN ISO 5079:2020 method to evaluate the improvements between pure PVDF & PVDF- Bi_2O_3/CeO_2, where the tensile strength increased with the addition of the high Z element (Bi_2O_3/CeO_2) in the PVDF.

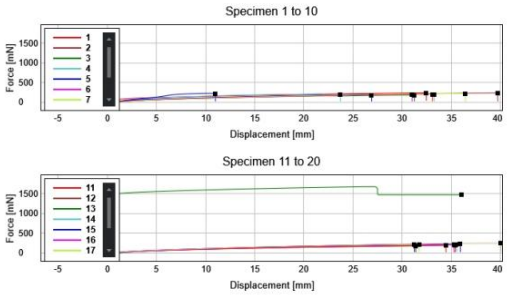


Figure 18 Tensile test for Pure PVDF Yarn

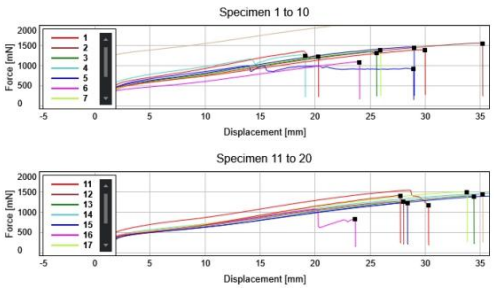


Figure 19 Tensile test for PVDF Bi_2O_3/CEO_2 Yarn

The PVDF-Bi_2O_3/CeO_2 composite exhibits a significantly enhanced tensile strength of 1.27 N compared to pure PVDF, which has a tensile strength of only 0.27 N. However, the tensile displacement at break for pure PVDF averages 32.3, while the PVDF-Bi_2O_3/CeO_2 composite fails at 29.6 mm, indicating reduced elasticity.

4 CONCLUSION

The incorporation of high-Z elements (Bi_2O_3 and CeO_2) into PVDF demonstrates excellent X-ray shielding performance while enhancing the polymer's β-phase mechanical properties, showing promising potential for functional protective textiles. The experimental results confirm that these high-Z elements contribute to X-ray shielding capabilities.

However, further optimization is required to achieve consistent and reproducible yarn for a textile-scale application, which necessitates additional investigation, including the development of more stable solvents in composite matrix dispersion that ensures uniformity, and the prevention of metal oxide nanoparticles from becoming trapped in the electrospinning process. The optimization of material ratios, particularly the balance between PVDF and Bi_2O_3 and CeO_2, underscores the importance of further research on optimal compositions to maximize shielding effectiveness while preserving textile and mechanical flexibility [7]. This dual-functional approach, which combines radiation protection with piezoelectric properties through enhanced β-phase crystallinity, represents a significant advancement toward next-generation smart protective textiles. Future work will continue to develop this composite by improving yarn formation and incorporating more textile components [6].

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